

HDI Predictor Documentation

Solution Requirements

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Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Input Handling	Users can enter life expectancy, mean years of schooling, expected years of schooling, and GNI per capita (PPP) through a web form	High
2	Input Validation	The system validates that all fields are numeric and within realistic ranges (e.g., life expectancy 0-100 years) before prediction	High
3	Prediction Engine	The system classifies the input into one of four HDI tiers — Very High, High, Medium, Low — using a trained RandomForestClassifier	High
4	Score Estimation	The system returns the underlying HDI numeric estimate alongside the categorical tier	Medium
5	Explainability	The system surfaces which input feature most influenced the prediction (feature importance)	Medium
6	Result Display	Results (tier, score, feature summary) are rendered back to the user on the same page without a full reload	High
7	Model Persistence	The trained model is serialized (model.pkl) and loaded once at application startup for	High

		fast repeated inference	
8	Deployment Packaging	The full application (Flask app + model + dependencies) is containerized with Docker for reproducible deployment	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction requests should return quickly enough for interactive use	Response in < 1 second per request
2	Scalability	The Flask app and model loading should support multiple sequential users without restart	Handles 50+ sequential requests without memory growth
3	Accuracy	The classifier must meet a high accuracy bar on held-out test data	≥ 95% test-set accuracy (achieved: 97.6%)
4	Reliability & Availability	The containerized app should start reliably and stay available during a demo session	Container starts successfully on first run
5	Usability & Accessibility	The form should be simple enough for a non-technical policymaker to use without instructions	Task completed by a first-time user in under 2 minutes
6	Portability	The application should run identically across different machines via Docker	Runs with a single <code>docker run</code> command, no manual dependency setup